Zyndio Build: PRD and Technical Plan

Oct 22, 2026 · @Aditya Singh

Read this first

Zyndio has signed and paid, so we build now, but the contractual Kick-off Date has not started yet because the Zyndio Inputs are still outstanding. Every deadline in this plan hangs off that date, so getting the inputs in this week is the first engineering task, not an admin chore.

Where the deal stands (as of Oct 22)

Item	Status	Where it comes from
Agreement	Revised version sent Oct 10 (Delaware law, New York arbitration held virtually, C3 fee ceiling). Brett says he signed on Oct 17 "via the DocuSign link", but we sent it through Zoho Sign. Signed copy and our countersignature still to be confirmed.	LinkedIn thread
Onboarding and build fee	USD 500 received through Skydo on Oct 21, after Razorpay failed three times	LinkedIn thread
Zyndio Inputs (Schedule A, Part 3)	Not received as far as the files show	Agreement
Kick-off Date	Not started. It is the latest of: last signature, fee receipt, Zyndio Inputs	Clause 1
Tier	Core, USD 399 per month at launch rate, 1,000 Active Patients included	Schedule B
Outcomes Partner	DEXASCAN, 400+ US testing locations	Press release
Zyndio technical counterpart	Matthew Trainer, CTO. Not yet formally named as IT lead	Press release

The clocks that drive this plan

- **Store Submission:** target 30 days, hard limit 45 days after Kick-off (clause 3.2). Missing 45 days lets Zyndio terminate and take the fee back (clause 6.5). Delays caused by Zyndio or by Apple and Google extend the window day for day (clause 3.3), so we log every wait.
- **Billing Start:** the earlier of store approval or 30 days after first Store Submission.
- **C3 quote:** fixed written quote within 10 business days of the Outcomes Specification being agreed.
- **Press release, Nov 2:** the text itself says the integrations are not being announced as operational. Brett still told us on Oct 10 that "we have 3 weeks to get this done", so plan for a branded demo build in his hands before Nov 2.

Five things the team must not get wrong

1. **Fictional data only** until the BAA is signed and Compliance Sign-off is given. No real patient data in any environment, not even one test record.
2. **Nostavia stays invisible.** No Nostavia name, domain, header, ID prefix, email sender or certificate anywhere a Zyndio patient or DEXASCAN can see it. We never contact DEXASCAN directly.
3. **Store Submission inside 45 days of Kick-off.** This is the one date with a refund attached.
4. **Decision support only.** No diagnosis, no dosing, and no narrative reaches a partner unless a Zyndio clinician wrote or approved it.
5. **C3 is capped** at USD 750 one-time and USD 99 per month, whatever DEXASCAN's data turns out to look like. C1 has to land a route that fits inside that cap.

Sources used and gaps

The signed agreement is the source of truth for scope and dates; Zyndio's strategy deck is the source of truth for what the outcomes loop should produce. Where they disagree, the agreement wins (for example, the deck leaves open whether we ingest DEXASCAN data directly, while the agreement routes it through a Zyndio Endpoint).

Source	What it gave us	What is missing
Platform Licence Agreement, revised Oct 2026	Scope (Schedule A), fees and tiers (B), outcomes workstream C1 to C3 (C), HIPAA and security (D), SLAs (E), discretion clause 15.2	Zyndio party block is blank: legal entity, state, address, notice email

Source	What it gave us	What is missing
Zyndio strategy deck, Clinical Data Return and Outcomes Intelligence Strategy (shared Oct 1)	Biomarker list, wearable signals, GLP-1 treatment context, derived outcomes, 25 to 35 field payload, governance, the seven technical decisions, sandbox success criteria	No data formats, volumes or DEXASCAN specifics
Zyndio and DEXASCAN press release (goes out Nov 2)	Outcomes Partner is DEXASCAN, 400+ locations, DEXA outputs (body fat %, lean mass, regional, visceral fat estimate, bone), some sites offer RMR and VO2 max, 90/180/365 day milestones, Zyndio CTO is Matthew Trainer	How DEXASCAN delivers results (API, file, PDF), which scanner vendors its network uses
LinkedIn thread with Brett, Sep 4 to Oct 21	Commercial history, the C3 ceiling we committed to on Oct 7, governing law change, demo expectations, payment status	Brett's email referenced on Sep 17, our September proposal, the two images he shared on Sep 16 and 17
Meeting transcripts (Sep 15 intro call, Oct 10 wrap-up call)	Not reviewed	Gmail, Google Drive and Google Calendar are connected to the account but switched off for this chat, so the transcripts could not be pulled. This doc will be updated once they are enabled

Assumption on our stack. Our current architecture docs are not in the files, so this plan is written stack-neutral. Where it names a pattern (queues, object storage, OAuth client credentials), the team uses our existing Platform equivalent rather than introducing something new.

Product requirements

We are giving Zyndio a branded patient app and physician portal now, and the plumbing for a closed loop with DEXASCAN that shows the quality of a patient's weight loss between scans, not just pounds lost.

Problem. Zyndio runs physician-led preventive and metabolic care in the US, with GLP-1 treatment as a lead use case. DEXASCAN measures body composition at 400+ locations, and its customers keep asking what to do after the scan. Zyndio needs one patient record that joins scans, labs, wearables and treatment, so its physicians can act on it and so DEXASCAN gets back a clean, clinician-reviewed outcome at 90, 180 and 365 days. Zyndio has no engineering platform for this today; we are that platform, unbranded.

Users

User	Who	What they need from us
Patient	Eligible US adults in a Zyndio programme, many on GLP-1 treatment	Easy sign-in from Zyndio, lab upload, wearable connection, food logging, clear trends, a companion that knows its limits, reminders for the next scan or lab
Zyndio physician and care team	Licensed providers and care coordinators	Roster with flags, one timeline per patient, pre-follow-up brief, treatment context, an outcomes review queue they approve from
Zyndio IT	Matthew Trainer's team	Enrolment and status APIs, webhooks, treatment data API, sandbox, logs, usage
Outcomes Partner	DEXASCAN, reached only through Zyndio	A 25 to 35 field milestone payload, never the whole record
Nostavia support and engineering	Us	Tenant config, monitoring, unmatched records, export and deletion tooling

Goals and how we measure them

Goal	Measure	Target
Ship the branded app	Store Submission date on iOS and Android	Inside 30 days of Kick-off, never past 45
Clean acceptance	Material non-conformities Zyndio reports in its 5 business day review	Zero open at submission
Prove the closed loop	C2 sandbox round trip	Passes all four success criteria in Schedule C
Reliable platform	Monthly availability of APIs and physician portal	99.5% or better (Schedule E)
Accurate labs	Field-level accuracy on our labelled lab fixture set (analyte, value, unit, date)	Internal target 98% on the priority analytes, every miss visible to the clinician
Patient adoption after go-live	Enrolled to first sign-in, wearable connected, baseline lab uploaded	Set with Zyndio at go-live; we instrument all three from day one

Non-goals for this agreement

- EHR or telehealth platform integration, direct Quest or other lab connections, CGM input, device integrations beyond Apple Health and Health Connect (Schedule A, Part 2).
- Prescribing, dosing or diagnostic logic of any kind (clause 9.1).
- The production DEXASCAN connector and live outbound delivery until a C3 Change Order is signed.
- Cohort and population analytics. We design the data model so they are possible later (deck slide 19), but they are quoted separately.
- Any Nostavia-branded surface, email, page or identifier.

Scope by phase

The work splits into four phases, and only the first two are paid for today. Each phase has a hard entry gate; nobody starts a phase early by borrowing real data or skipping a signature.

1. **Phase 0, pre-kick-off (now to Kick-off).** Tenant set-up, white-label build pipeline, synthetic data, inputs checklist to Zyndio. Gate out: Zyndio Inputs received, signed agreement confirmed.

2. **Phase 1, platform build (Kick-off to Store Submission).** Everything in Schedule A, Part 1 below, plus the C1 architecture session. Gate out: Zyndio accepts the fictional-data build (or 5 business days pass with no report), then Store Submission.
3. **Phase 2, outcomes sandbox (after Store Submission).** C2 round trip on fictional data through a test Zyndio Endpoint. Gate out: the four Schedule C success criteria pass, Outcomes Specification signed, C3 quote sent within 10 business days.
4. **Phase 3, production outcomes (C3).** Production connector and live delivery. Gate in, all required: C3 Change Order signed, BAA executed, Compliance Sign-off, Zyndio's own agreement with DEXASCAN, patient authorisations captured.
5. **Phase 4, live operations.** Real patients, metering and billing, SLAs, 90/180/365 day milestones. Gate in: BAA and Compliance Sign-off. This can start before Phase 3 for the app and portal alone.

Phase 1 deliverables (Schedule A, Part 1)

Items marked "available today" still need configuration and testing work for Zyndio; nothing here is zero effort.

Deliverable	Contract status	Engineering work for Zyndio	Lead
Zyndio-branded iOS and Android app	Built for Zyndio	White-label build, Zyndio sign-in, consent and onboarding, store listing, signing under Zyndio's developer accounts	Mobile
Wearable data	Built for Zyndio	Apple Health and Health Connect reads, on-device daily aggregation, normalisation into one format	Mobile + Backend
zyndio.com integration	Built for Zyndio	Enrolment, account status and programme data APIs and webhooks. Zyndio's IT changes its own site and backend	Backend
Lab report upload and biomarker extraction	Available today	Tenant config, report-to-patient identity check, LOINC map review for the deck's priority analytes	Data

Deliverable	Contract status	Engineering work for Zyndio	Lead
Biomarker interpretation	Available today, configured	Load Zyndio's ranges when supplied (defaults until then, demo only), alert thresholds, trend views	Data + Clinical
Physician portal	Available today, configured	Roles mapped to Zyndio's team, roster flags, pre-follow-up brief template, export, Zyndio branding	Web
Food logging and AI companion	Available today, configured	Zyndio boundaries, companion name and tone, routing clinical questions to the care team	AI
Safety behaviours	Available today	Emergency language screening, insufficient-data verdicts, physician flags wired to Zyndio's escalation contacts	AI + Clinical
Modules	Available today	Core tier: five core modules plus two standard modules of Zyndio's choice, activated from the Partner Portal	Backend

Not in scope without a Change Order (Schedule A, Part 2): Quest or other direct lab connections, CGM input, EHR or telehealth platform integration, device integrations beyond Apple Health and Health Connect, the production DEXASCAN connector and outbound delivery (C3), custom features beyond Part 1. If anyone asks us to "just add" one of these, it goes to Aditya for a written quote within 10 business days (clause 3.5).

User journeys and functional requirements

The core journey is DEXASCAN's "From Scan to Plan": baseline, physician plan, monitoring, re-scan at 90, 180 and 365 days, and a reviewed outcome back to the partner. Every requirement below traces to a step in it.

Core journey (one patient, end to end)

1. Zyndio enrols the patient on zyndio.com. Zyndio's backend calls our enrolment API and the patient gets a Zyndio-branded invite.

2. Patient installs the Zyndio app, signs in, accepts Zyndio's terms and privacy notice, and separately authorises sharing outcomes with the body composition partner.
3. Patient connects Apple Health or Health Connect and uploads a baseline lab report.
4. Patient gets a baseline DEXA at a DEXASCAN location. DEXASCAN sends the result to the Zyndio Endpoint, which forwards it to us, and we bind it to the patient.
5. Zyndio physician reviews the combined baseline, starts treatment, and the treatment context (medication, dose, start date) reaches us by API or portal entry.
6. Ongoing: wearable sync, food logging, companion questions, lab uploads, physician flags.
7. Before each milestone the patient is reminded to book the scan and labs.
8. Follow-up scan and labs arrive. The outcomes engine builds the milestone record; a Zyndio clinician reviews and approves it; it is delivered to Zyndio, which passes it to DEXASCAN.

Priorities. P0 is needed for acceptance and Store Submission. P1 is needed before the first real patient (BAA and Compliance Sign-off). P2 belongs to the outcomes phases.

Patient app

ID	Requirement	Priority
APP-01	Sign-in from a Zyndio enrolment invite (magic link or one-time code). No open self-sign-up unless Zyndio switches it on	P0
APP-02	Consent and onboarding: Zyndio terms, privacy notice and consent text, each stored with version and timestamp	P0
APP-03	Separate, revocable authorisation to share outcomes with the body composition partner. Revoking it stops future deliveries	P1
APP-04	Profile: name, date of birth, sex at birth (drives ranges), height (drives BMI), unit preference	P0
APP-05	Lab upload by camera, image or PDF, multi-page. Shows extracted values, flags a report whose name or date of birth does not match the account	P0
APP-06	Biomarker views: trend graph, single-marker history, change since baseline, Zyndio ranges, bounded alert wording	P0

ID	Requirement	Priority
APP-07	Connect, view status of and disconnect Apple Health or Health Connect, with per-type permission prompts	P0
APP-08	Food logging by photo or text	P0
APP-09	AI companion inside Zyndio's boundaries. Clinical questions route to the care team; emergency language shows emergency guidance and alerts Zyndio's escalation contact	P0
APP-10	Push notifications with no health information in the visible text	P0
APP-11	In-app account deletion request, routed to Zyndio, with clear wording on records Zyndio must legally keep	P0
APP-12	No Nostavia string anywhere: about screen, errors, emails, deep links, crash dialogs	P0
APP-13	Body composition view once scans arrive: baseline against follow-up, fat, lean and visceral trends	P2
APP-14	Milestone reminders for the next scan and lab (fuller automation if Zyndio adds the Re-Test Recall module)	P2

Physician portal

ID	Requirement	Priority
POR-01	Sign-in with mandatory MFA. Roles: Physician, Care coordinator, Admin, Read-only, mapped to Zyndio's team	P0
POR-02	Patient roster with flags: alerts, clinical context flags, missing data, unanswered companion escalations	P0
POR-03	Combined timeline: labs, wearables, scans, food, companion escalations, treatment events	P0
POR-04	Pre-follow-up brief per patient	P0

ID	Requirement	Priority
POR-05	Range and protocol management, versioned, physician-approved before live use (clause 4.2)	P0
POR-06	Data export per patient	P0
POR-07	Treatment context entry (medication, formulation, dose changes, status, tolerability) for when Zyndio's backend does not send it by API	P1
POR-08	Audit log view for Zyndio admins: who viewed or changed which record	P1
POR-09	Outcomes review queue: computed milestone records, edit or write the narrative, approve, hold or send back	P2
POR-10	Unmatched partner records queue: Zyndio staff bind an incoming scan to the right patient	P2

Zyndio backend and zyndio.com integration

ID	Requirement	Priority
INT-01	Enrol, update and deactivate a patient by API, keyed on Zyndio's own patient ID	P0
INT-02	Account status webhooks: enrolled, activated (first sign-in), deactivated, consent changed	P0
INT-03	Programme data read API: latest biomarkers, wearable summary, flags	P0
INT-04	Treatment events API: medication start, dose change, status change, tolerability note	P1
INT-05	Partner measurement ingestion API, called by the Zyndio Endpoint	P2
INT-06	Outcomes webhook to Zyndio plus retrieval API for reconciliation	P2

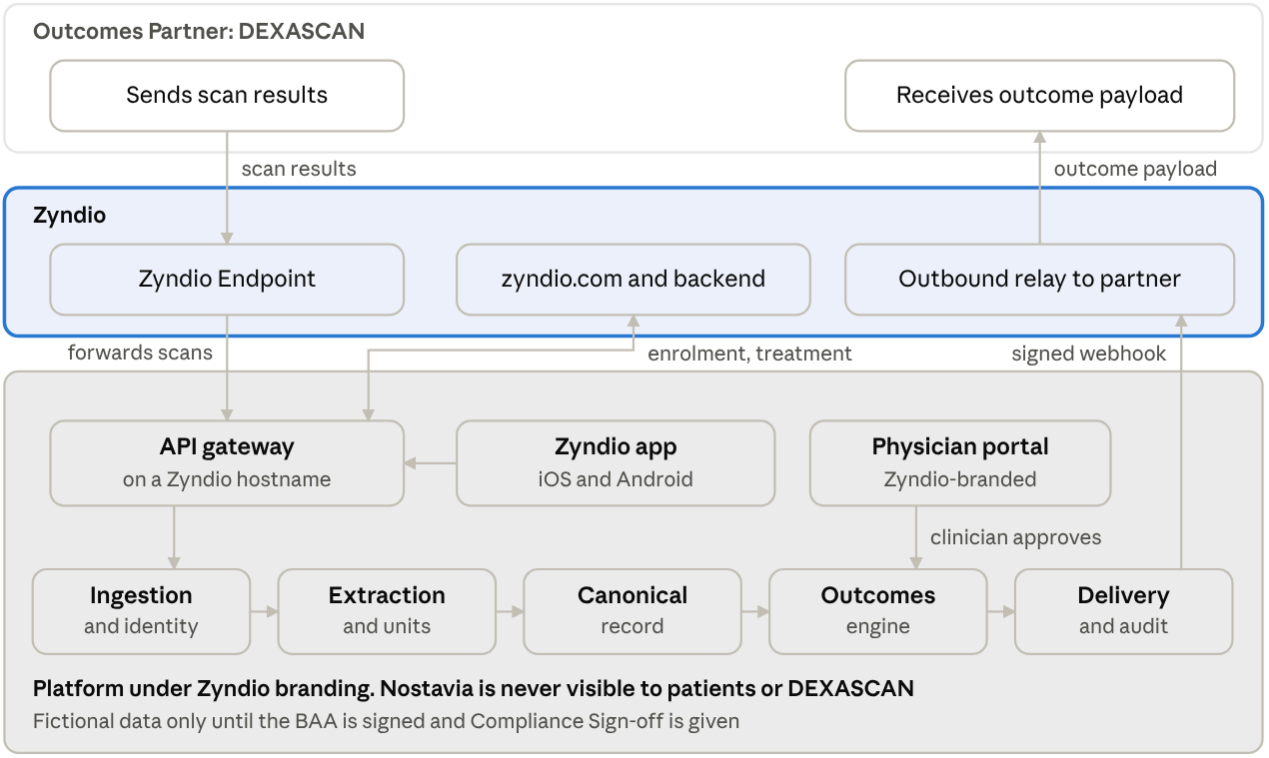
Partner Portal and Nostavia operations

ID	Requirement	Priority
PP-01	Tier and module changes under clause 5.8, with pro-rating data for invoicing	P0
PP-02	Active Patients per calendar month, with the reason each patient counted (sign-in or data processed)	P1
PP-03	API credentials, webhook endpoints, secret rotation, delivery log	P0 for INT, P2 for outcomes
OPS-01	Patient data export in an agreed machine-readable format within 30 days of a termination request	P1
OPS-02	Patient data deletion with a written deletion confirmation, honouring legal holds	P1
OPS-03	Security incident workflow that lets us notify Zyndio within 72 hours of confirming an incident	P1

System architecture

DEXASCAN only ever exchanges data with Zyndio, and every surface the platform exposes carries Zyndio's name. That is the shape clause 15.2 forces on us, and it is also the easiest one to audit.

DEXASCAN only ever talks to Zyndio; the platform behind it is never named



system architecture · 3 zones, 1 pipeline

Data enters three ways (the app, Zyndio's backend, the Zyndio Endpoint) through one gateway on a Zyndio hostname, runs through one pipeline into one record, and leaves only as a clinician-approved outcome sent back to Zyndio.

Components

Component	Responsibility	Status
API gateway	Auth, rate limits, request IDs, Zyndio hostname and certificate	Existing, add custom domain
Identity and tenant	Platform IDs, identity links, partner tokens, consent records, tenant config	Existing, extend for partner IDs and tokens
Ingestion	Lab documents, wearable daily uploads, partner scans, treatment events	Existing for labs; scans and treatment events are new
Extraction and units	Lab extraction, LOINC mapping, unit normalisation, confidence	Available today, configure for Zyndio

Component	Responsibility	Status
Canonical record	FHIR-aligned observations, documents, provenance, versions	Existing, extend for body composition and treatment
Interpretation and safety	Ranges, flags, emergency screening, companion boundaries	Available today, configure for Zyndio
Outcomes engine	Milestones, deltas, trends, completeness, clinical flags, payload build	New, for C2
Review queue	Clinician approval in the physician portal	New, for C2
Delivery and audit	Signed webhooks, retries, dead letter, delivery log	New for outcomes; reuse our webhook sender where one exists
Metering	Active Patient counting per month	New or extend existing usage job
Partner Portal	Tier and modules, keys, webhook config, usage, delivery log	Existing, extend

Environments

Environment	Data	Used by
Dev	Synthetic	Our engineers
Staging	Synthetic plus golden fixtures	QA, the acceptance build
Sandbox	Fictional only	Zyndio IT for integration and C2
Production	Real patient data only after BAA and Compliance Sign-off; US region, backups in US region	Zyndio, its patients

The Bespoke tier's option of running the backend inside Zyndio's own cloud account is not part of Core, so the plan assumes our managed US hosting.

Data model and identity mapping

One internal patient ID ties everything together and never leaves the platform; every outside party sees only its own identifier or an opaque token. This answers the deck's question on mapping partner IDs "without exposing unnecessary identifiers" and keeps us invisible under clause 15.2.

Canonical record. Store observations in a FHIR R4-aligned internal model (Patient, Observation, MedicationStatement, DocumentReference, Provenance) with LOINC codes and UCUM units. We do not expose FHIR to the partner; the outbound payload is a flat, versioned JSON document built from this record. Proposal for C1: this model is the canonical longitudinal record for the workflow.

Identifiers

Identifier	Issued by	Where it appears	Rule
platform_patient_id (UUID)	Us	Internal only	Never in an API response, payload, URL, log line visible to Zyndio, or push
zyndio_patient_id	Zyndio	Enrolment API, treatment API, webhooks to Zyndio	Zyndio's key for every server-to-server call
partner_patient_id	DEXASCAN	Inbound scans, outbound outcome payload	Stored as received, echoed back so DEXASCAN can match
zyndio_patient_token	Us, under Zyndio's name	Outbound outcome payload	HMAC-SHA256 of platform_patient_id with a per-partner secret. Stable per partner, useless elsewhere, rotatable

Binding rule for inbound scans. Zyndio owns patient matching. The Zyndio Endpoint must send zyndio_patient_id with every scan (from the DEXASCAN booking reference Zyndio issues). We never fuzzy-match on name or date of birth. A scan without a known zyndio_patient_id goes to the unmatched queue (POR-10) and raises a measurement.unmatched webhook. Once bound, partner_patient_id is stored against the patient so later scans bind automatically.

Entities

Entity	Key fields	Notes
Tenant	tenant_id, tier, modules, branding, ranges_version	Zyndio is one tenant; all queries are tenant-scoped
Patient	platform_patient_id, tenant_id, demographics needed for ranges	Minimum necessary demographics only
IdentityLink	platform_patient_id, issuer, external_id, linked_at, linked_by	One row per outside identifier; unique per issuer
Consent	patient, type (terms, privacy, partner_disclosure, wearable), version, granted_at, revoked_at	Partner disclosure checked at delivery time, not only at compute time
SourceDocument	id, patient, kind (lab_report, dexa_report), storage_uri, sha256, received_at, extraction_status, extractor_version	Original kept encrypted; every value traces back to it
Observation	id, patient, loinc, value, unit (UCUM), original_value, original_unit, effective_at, source_type, source_ref, confidence, status, version	status: preliminary, final, amended, entered-in-error. Never overwrite; supersede
BodyCompositionScan	id, patient, scan_date, device_vendor, device_model, site_id, partner_scan_id, partner_version, measures	Measures: total mass, fat mass, body fat %, lean mass, regional lean, visceral fat, bone metrics, whatever DEXASCAN supplies
WearableDaily	patient, date, metric, value, unit, source_app, source_device, coverage	Daily aggregates, not raw samples (see pipeline section)
TreatmentCourse	patient, medication, formulation, start_date, status	status: active, paused, discontinued, completed. Supplied by Zyndio
TreatmentEvent	course, type (dose_change, status_change, tolerability, clinician_change), value, effective_at, entered_by	Dose stored as recorded; we never suggest one

Entity	Key fields	Notes
Milestone	patient, type (baseline, d90, d180, d365, adhoc), anchor_date, window_start, window_end, status	Anchor is treatment start unless Zyndio says otherwise
OutcomeRecord	id, patient, milestone, baseline_refs, followup_refs, computed fields, rules_version, narrative, state, approved_by, approved_at, record_version	Immutable once delivered; amendments create a new record_version
Delivery	outcome_record, destination, event_id, attempt, status, response_code, payload_sha256, sent_at	Full audit of what left, when, and to whom
AuditEvent	actor, action, resource, at, ip, request_id	Append-only, covers reads of patient records too
UsageMonth	tenant, month, platform_patient_id, reason	Feeds Active Patient billing

Design rules. Every value keeps its original value, unit and source. Nothing is deleted in place; corrections supersede. All timestamps are stored in UTC with the source time zone kept. The same normalised record must support cohort analysis later, so no outcome logic lives only in the payload builder.

Lab ingestion, wearables and normalisation

Three inputs feed one record: lab reports the patient uploads, wearable data from the phone, and DEXA results from the partner. Each keeps its original value, unit and source so any number in an outcome can be traced back to a page or a device.

Lab report pipeline

1. **Receive.** File type and size check, malware scan, encrypted storage, SourceDocument row with SHA-256. A repeat upload of the same file is detected and not processed twice.
2. **Extract.** Our existing engine pulls analyte, panel, value, unit, printed reference range, collection date, lab name, and the name and date of birth printed on the report.
3. **Identity check.** Printed name or date of birth that does not match the account puts the report on hold and asks the patient to confirm. Physicians see the hold.

4. **Map.** Analyte name to LOINC through a reviewed mapping table. Unknown analytes are stored as unmapped and never feed outcomes.
5. **Normalise.** Convert to the canonical unit below; keep the original value and unit and the factor used.
6. **Validate.** Physiological plausibility limits, duplicate detection (same analyte, date and value), fasting status where the report states it.
7. **Score.** Per-field confidence. Below threshold, the value is marked "needs review" and shown to clinicians next to the source snippet.
8. **Persist and interpret.** Observations written, Zyndio ranges applied, flags raised, events emitted.

All model calls on real patient data go only to AI providers that have signed a BAA and contractually do not train on our data (clause 8.6). Until Compliance Sign-off, the pipeline only ever sees fictional reports.

Canonical units for the priority analytes

US conventional units are canonical because Zyndio and DEXASCAN are US-facing. Every factor below gets a unit test, and code review checks it against a real report that prints both units.

Analyte	Canonical unit	Common alternate	Conversion to canonical
HbA1c	% (NGSP)	mmol/mol (IFCC)	% = mmol/mol / 10.929 + 2.15
Glucose (fasting)	mg/dL	mmol/L	x 18.016
Insulin (fasting)	uIU/mL	pmol/L	/ 6.0 (assay dependent; store the factor used)
Total, LDL, HDL, non-HDL cholesterol	mg/dL	mmol/L	x 38.67
Triglycerides	mg/dL	mmol/L	x 88.57
ApoB	mg/dL	g/L	x 100
Lp(a)	as reported	mg/dL or nmol/L	Do not convert. The relationship is isoform dependent, so trend only within the same unit
hsCRP	mg/L	mg/dL	x 10
ALT, AST, GGT	U/L	ukat/L	x 60

Analyte	Canonical unit	Common alternate	Conversion to canonical
Creatinine	mg/dL	umol/L	/ 88.42
eGFR	mL/min/1.73m2	none	Take as reported with the equation if printed; do not recompute
25-OH vitamin D	ng/mL	nmol/L	/ 2.496
Ferritin	ng/mL	ug/L	1:1
Testosterone, total	ng/dL	nmol/L	x 28.84
Estradiol	pg/mL	pmol/L	/ 3.671
Free T4	ng/dL	pmol/L	/ 12.87
Free T3	pg/mL	pmol/L	/ 1.536
IGF-1	ng/mL	nmol/L	x 7.649
Weight	kg (stored), lb (displayed)	lb	x 0.45359237

HOMA-IR is calculated only when fasting glucose and fasting insulin come from the same draw, and only where Zyndio's protocol says it is clinically appropriate:

```
\text{HOMA-IR} = \frac{\text{fasting insulin (uIU/mL)} \times \text{fasting glucose (mg/dL)}}{405}
```

Wearables

	iOS (Apple Health)	Android (Health Connect)
Types read	Steps, exercise minutes, active energy, resting heart rate, HRV (SDNN), sleep analysis, body mass, BMI, blood pressure, VO2 max, oxygen saturation	Steps, exercise sessions, resting heart rate, HRV (RMSSD), sleep sessions, weight, blood pressure, VO2 max, oxygen saturation
Sync	Observer queries with background delivery, anchored queries for increments, statistics collection queries for daily totals	Periodic background worker; needs the background read permission

	iOS (Apple Health)	Android (Health Connect)
History on connect	Backfill 90 days so a pre-treatment baseline exists	Default access only reaches 30 days before the first grant; older data needs the separate history permission. Request it so the 90-day backfill works
Store review	Usage descriptions for every type; no health data used for advertising	Health apps declaration in Play Console must match the manifest permissions exactly, or the build is rejected

Rules for wearable data:

- **Aggregate on the device.** Upload one value per metric per day (sleep per night, assigned to the wake date) with coverage and source device. No raw heartbeat or step streams reach our servers. This is less data to protect and matches the deck's "no raw event firehose" rule.
- **Never mix HRV methods.** iOS gives SDNN, Health Connect gives RMSSD. They are stored as separate metrics and trended only against themselves.
- **One source per metric per day.** When a watch and a phone both report steps, use the configured priority source; never add them.
- **Uploads are idempotent** on patient, date, metric and source, so a retry or a re-sync never double counts.

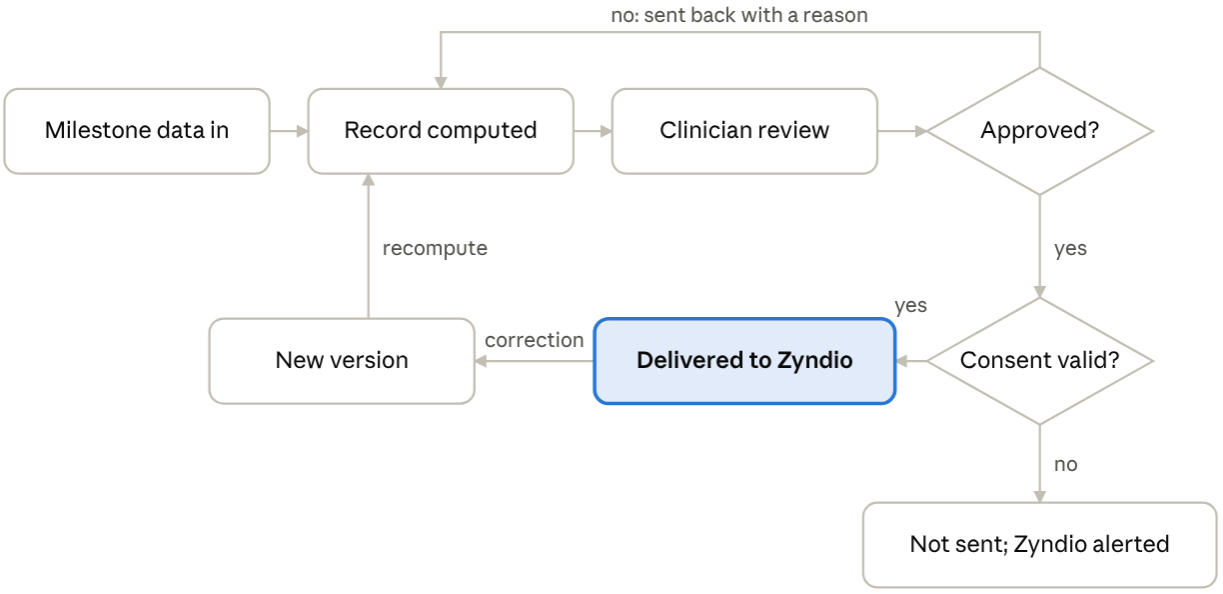
DEXA results

- Store device vendor, model and software version with every scan. Visceral fat estimates and regional values are not comparable across vendors or software versions, so a change of device sets that measure's trend to "insufficient data" with a method-change reason.
- Pin the definition of lean mass in the Outcomes Specification. DEXA reports usually separate lean tissue from bone mineral content. The deck's own example shows this: 82 lb fat plus 137 lb lean is 219 lb of a 228 lb total, the remainder being bone and rounding.
- Accept structured data first (JSON, CSV or HL7/FHIR from DEXASCAN through the Zyndio Endpoint). If DEXASCAN can only give PDF reports, route them through the same document pipeline as labs, scoped to the scanner vendors its network actually uses.

Outcomes engine

The engine turns a baseline and a follow-up into one milestone record of 32 to 35 fields, using rules a clinician can read. It computes and compares; it never concludes. Zyndio's clinicians decide what the numbers mean (deck slide 17, clause 9.1).

Nothing leaves until a clinician approves and consent is still valid



outcome record lifecycle · 2 gates

A record is built when the follow-up scan is bound or the milestone window closes, whichever comes first. A record that has already been delivered is never edited: a correction (a late lab, a DEXASCAN rescan version) produces a new record_version that goes through review again.

Derived outcomes (Schedule C)

Output	Rule
% total body-weight change	(follow-up weight minus baseline weight) divided by baseline weight, times 100, reported with elapsed treatment days
Biomarker delta	Absolute and percentage change, with both timestamps and the canonical unit. Only computed when both values share a comparable unit and method
Trend direction	improving, stable, worsening or insufficient_data, per the rules below, always with a trend_reason

Output	Rule
Data completeness	Per domain (scan, labs, wearables, treatment): expected items present in the window divided by expected items. The expected set is defined in the Outcomes Specification
Lean-mass flag	none, watch or flagged, from the scan's lean mass change against Zyndio's threshold
Clinical context flag	Raised when a value crosses a Zyndio critical threshold, a safety marker (ALT, AST, GGT, eGFR) worsens, the lean-mass flag is "flagged", an adverse effect is recorded, treatment is paused or stopped, or any input is still in "needs review". A flagged record cannot be approved until a clinician clears the flag

Trend rules

1. For each measure, Zyndio's physician lead sets a direction of benefit and a minimum meaningful change (absolute or relative).
2. **insufficient_data** if either value is missing, falls outside the milestone window, or the method changed between the two (different scanner vendor or software, LDL calculated against direct, HRV SDNN against RMSSD, Lp(a) in different units).
3. Otherwise **improving** if the change runs in the direction of benefit by at least the minimum, **worsening** if it runs against it by at least the minimum, and **stable** if it is smaller.
4. Each record stores the rules_version it was computed with. Changing a threshold creates a new rules version; it never silently rewrites delivered records.

Default direction of benefit, for the physician lead to confirm before live use

Direction	Measures
Lower is better	HbA1c, fasting glucose, fasting insulin, HOMA-IR, triglycerides, ApoB, LDL-C, non-HDL-C, hsCRP, blood pressure, resting heart rate, fat mass, visceral fat, weight in a weight-management programme
Higher is better	HDL-C, HRV (within one method), VO2 max, lean mass, steps, active minutes

Direction	Measures
Judged against range, not direction	ALT, AST, GGT, creatinine, eGFR, thyroid, sex hormones, vitamin D, ferritin, IGF-1. These drive the clinical context flag rather than an improving or worsening label

The minimum meaningful change values are not in any file we have, and they are a clinical call, so the engine ships with them as required configuration and refuses to compute a trend until Zyndio's physician lead has set them.

Narrative. The portal pre-fills a template summary from the numbers (fixed wording, not free AI text) for the clinician to edit. Only text a Zyndio clinician has written or approved is ever sent, matching the deck's rule against unreviewed AI conclusions.

API and webhook contracts

Webhook-first, API for reconciliation, exactly as the deck asks; we own delivery to Zyndio, and Zyndio owns delivery to DEXASCAN. That split is our proposed answer to the deck's question on who owns the outbound partner API, and it is the starting position in Schedule C.

Conventions for every endpoint

- **Host:** a Zyndio-branded hostname (for example platform.zyndio.com) pointed by CNAME at a neutral edge hostname, with the TLS certificate issued for Zyndio's name. No Nostavia string in hostnames, certificates, headers, error bodies or ID prefixes.
- **Auth:** OAuth 2.0 client credentials, 15-minute bearer tokens, scopes per capability (patients:write, patients:read, treatment:write, measurements:write, outcomes:read, webhooks:manage). Separate clients for sandbox and production. IP allowlisting optional.
- **Format:** JSON, snake_case, ISO 8601 UTC timestamps, explicit units on every measure.
- **Idempotency:** every POST takes an Idempotency-Key header; replays within 7 days return the original result.
- **Errors:** one shape, `{"error":{"code","message","request_id"}}`, with stable codes.
- **Versioning:** major version in the path (/v1). Additive changes are non-breaking. A breaking change ships as /v2 with at least 90 days of overlap.
- **Pagination:** cursor based, with updated_since filters for reconciliation.

Endpoints

Method and path	Purpose	Priority
POST /v1/patients	Enrol a patient by zyndio_patient_id and send the Zyndio invite	P0
PATCH /v1/patients/{zyndio_patient_id}	Update contact details or deactivate	P0
GET /v1/patients/{zyndio_patient_id}	Account status, consent state, connected sources	P0
POST /v1/patients/{zyndio_patient_id}/invites	Resend the invite	P0
GET /v1/patients/{zyndio_patient_id}/summary	Programme data: latest biomarkers, wearable summary, open flags	P0
GET, POST, DELETE /v1/webhook-endpoints	Register endpoints, rotate secrets	P0
POST /v1/patients/{zyndio_patient_id}/treatment-events	Medication start, dose change, status change, tolerability	P1
GET /v1/patients/{zyndio_patient_id}/treatment	Current course and event history	P1
POST /v1/measurements/body-composition	Zyndio Endpoint forwards a structured scan	P2
POST /v1/measurements/body-composition/documents	Zyndio Endpoint forwards a scan report as PDF	P2
GET /v1/outcomes/{record_id}	One outcome record, latest version	P2
GET /v1/outcomes?updated_since=	Reconciliation feed	P2
POST /v1/outcomes/{record_id}/redeliver	Resend a delivered record	P2

Webhook events

Event	Fires when	Priority
patient.enrolled, patient.activated, patient.deactivated	Enrolment accepted, first sign-in, deactivation	P0

Event	Fires when	Priority
consent.updated	Patient grants or revokes any consent, including partner disclosure	P0
measurement.received, measurement.bound	Scan accepted, scan bound to a patient	P2
measurement.unmatched, measurement.rejected	No known patient, or failed validation (reason code included)	P2
outcome.approved	Clinician approved a milestone record; carries the payload	P2
outcome.amended, outcome.withdrawn	A delivered record was corrected (new record_version) or pulled back	P2

Webhook delivery rules

- Envelope: event_id, event_type, created_at, schema_version, data.
- Signature header: `X-Signature: t=<unix time>,v1=<hex HMAC-SHA256 of "t.body">` with a per-endpoint secret. Receivers reject anything older than 5 minutes.
- At-least-once delivery. Receivers dedupe on event_id and order on record_version, not arrival time.
- Retries on any non-2xx or no answer within 10 seconds: after 1 min, 5 min, 30 min, 2 h, 6 h, 12 h and 24 h. Then dead letter, alert our on-call, and show it in the Partner Portal delivery log.
- Every attempt is logged with payload hash, response code and timing (Delivery entity), which gives the deck's "auditability".

Proposed latency targets (our SLOs, not contract terms)

Flow	Target at p95
Structured scan received to bound	5 minutes
PDF scan or lab report to extracted values	30 minutes
Clinician approval to outcome.approved sent	15 minutes
Wearable daily upload to visible in portal	1 hour

Example: scan forwarded by the Zyndio Endpoint (fictional)

```
{
  "zyndio_patient_id": "ZP-104233",
  "partner_patient_id": "DX-88120",
  "partner_scan_id": "SCAN-55102",
  "partner_scan_version": 1,
  "scan_date": "2026-12-01",
  "site_id": "SITE-0421",
  "device": {"vendor": "VendorA", "model": "ModelX", "software_version":
"13.6"},
  "measures": {
    "total_mass": {"value": 228.0, "unit": "lb"},
    "fat_mass": {"value": 82.0, "unit": "lb"},
    "body_fat_pct": {"value": 36.0, "unit": "%"},
    "lean_mass": {"value": 137.0, "unit": "lb"},
    "visceral_fat_mass": {"value": 1450, "unit": "g"}
  }
}
```

Response: `202 Accepted` with `{"ingestion_id":"ing_7Qx2","status":"received"}`. Idempotency is keyed on partner_scan_id plus partner_scan_version, so DEXASCAN corrections arrive as a new version, never a silent overwrite.

Security, HIPAA and white-label discretion

The Zyndio production tenant stays locked for enrolment until Compliance Sign-off is recorded; that single switch is how we enforce "no real data before the BAA". The engineering lead and Aditya are the only two people who can flip it, and the flip is audit logged.

Compliance Sign-off checklist (Schedule D, Part 1)

- [ ] Data flow map covering intake, storage, extraction, AI calls, physician portal, export, the Zyndio Endpoint and the outcomes payload
- [ ] US-region production hosting and backups confirmed
- [ ] Subprocessor list with role and agreement status: cloud, database, object storage, AI model providers, email and SMS sender, push, error tracking. Any vendor without a BAA is removed from the path of identifiable data
- [ ] Role-based access, MFA and audit logging demonstrated on the staging build
- [ ] Security pack (architecture, encryption, incident response, vulnerability handling) written so Zyndio can share it under its own name (clause 8.9)
- [ ] In-app consent flows and privacy notice aligned to Zyndio's patient terms

- [ ] HIPAA roles confirmed with Zyndio's counsel and the BAA executed (Zyndio's form if it has one, otherwise ours)
- [ ] Regulatory classification written down: outputs are decision support for physician review, with the clinician-directed and informational split agreed

Security controls

Area	Requirement
Encryption	TLS 1.2 or higher in transit; AES-256 at rest with managed keys; direct identifiers (name, date of birth, email, phone) encrypted at field level
Access	Least privilege roles; MFA for every clinician and admin; production access only through approved, time-boxed, logged sessions; no shared accounts
Audit	Append-only log of every read and change of patient data, retained 6 years by default (confirm in the BAA)
Environments	Dev, staging and sandbox hold fictional data only. Production data is never copied down
Logging	No patient data in application logs, traces or crash reports. Log internal IDs and request IDs only
Mobile	Health data in Keychain or Keystore backed encrypted storage, nothing health-related in device logs, session timeout, optional biometric lock
Push and email	No health information in push text or email subject and body; the message only says there is something to view in the app
Third-party SDKs	No product analytics or ad SDKs in the app unless BAA-covered and approved by Zyndio
Resilience	Proposed internal targets: recovery point 1 hour, recovery time 8 hours, restore test before go-live
Vulnerabilities	Dependency and container scanning plus static analysis in CI; critical fixes within 7 days; external penetration test before the first real patient (recommended, not contractual)

Area	Requirement
Incidents	Written plan; Zyndio notified within 72 hours of confirming an incident (clause 8.8)
Subprocessor changes	30 days' notice to Zyndio before adding one that touches PHI (Schedule D, Part 3)
Certification claims	Our cloud provider is SOC 2 Type II certified; Nostavia itself is not. Nobody on the team says otherwise, in writing or on a call

Clinical boundary in code

- Every output is labelled and built as decision support. No feature suggests a dose, a diagnosis or a prescription.
- The companion answers inside Zyndio's boundaries and hands clinical questions to the care team.
- Every AI-assisted feature sits behind its own kill switch, so if a regulator objects to one we can switch it off for Zyndio within a day (clause 9.2) without a release.

Discretion checklist (clause 15.2)

Surface	Rule
App binary and store listing	Zyndio's bundle ID, app name, developer account and support contacts. Rename internal module namespaces in the white-label build so no Nostavia package name ships
Physician portal	Zyndio domain, page titles, favicon, footer and login emails
Email and SMS	Sent from Zyndio's domain. Zyndio IT adds the sender authentication DNS records
API and webhooks	Zyndio hostnames and certificates, neutral headers and user agent, server banners stripped, neutral ID prefixes (evt_, ing_, rec_)
Payloads	No Nostavia field names or identifiers; platform IDs never leave
Documentation	API docs and the security pack are issued under Zyndio's name

Surface	Rule
People	No Nostavia email, call or calendar invite to DEXASCAN or to patients without Zyndio's written approval

Non-functional requirements

The contract commits us to 99.5% monthly availability and a 4-hour response to Priority 1 issues at any hour, so we need a real on-call rota covering US daytime from day one of live use.

Service levels we have signed (Schedule E)

Commitment	Number	What engineering does about it
Availability, Platform APIs and physician portal	99.5% per month, about 3.6 hours of unplanned downtime in a 30-day month	Synthetic checks every minute from a US region; monthly availability report to Zyndio
Chronic failure	Below 99.0% two months running lets Zyndio exit with a refund	Any month under 99.5% gets a written post-mortem and a fix plan
Scheduled maintenance	48 hours' notice, outside 8am to 8pm US Eastern where practicable	That window is daytime in Lucknow: maintenance can run until 5:30 PM IST all year
Priority 1 (platform down or a patient-safety function failing)	Response within 4 hours, any time	Paging on-call rota, including weekends and Indian holidays
Priority 2 (major feature degraded, no workaround)	1 business day	Business days exclude Indian public holidays
Priority 3 (minor issue or question)	3 business days	Support hours overlap US Eastern mornings, which is IST evening

Commitment	Number	What engineering does about it
Included maintenance	Device and lab format changes, wearable connection upkeep, app compatibility within 30 days of a major iOS or Android release	Test on the iOS and Android betas every summer so the September releases are routine

Performance and scale

Area	Target
API reads	p95 under 500 ms
Physician portal roster and timeline	p95 under 2 seconds for a patient with a year of data
Launch capacity	1,000 Active Patients (the included amount)
Load tested to	10,000 Active Patients before go-live, so the first usage band is safe
Architecture headroom	50,000+ Active Patients without redesign (top usage band in Schedule B)

Metering and billing data

- An Active Patient is anyone who signs in to the Zyndio app or whose data the Platform processes in a calendar month. A nightly job writes one UsageMonth row per patient per month with the first qualifying reason.
- The Partner Portal shows the running count; month close produces the invoice figures. Usage above 1,000 is billed in arrears by band (USD 0.50, 0.35, 0.25).
- Background processing alone (for example a wearable sync) counts as "data processed". Zyndio should know that before go-live so the first usage invoice is not a surprise.

Observability

- Dashboards per flow: enrolment, lab extraction, wearable uploads, scan ingestion, outcome builds, webhook deliveries.
- Alerts on extraction failure rate, dead-lettered webhooks, unmatched scans older than 2 business days, and any outcome approved but not delivered within 1 hour.
- All of it built on internal IDs only, never patient identifiers.

Exit obligations we must be able to meet at any time (clause 6.6)

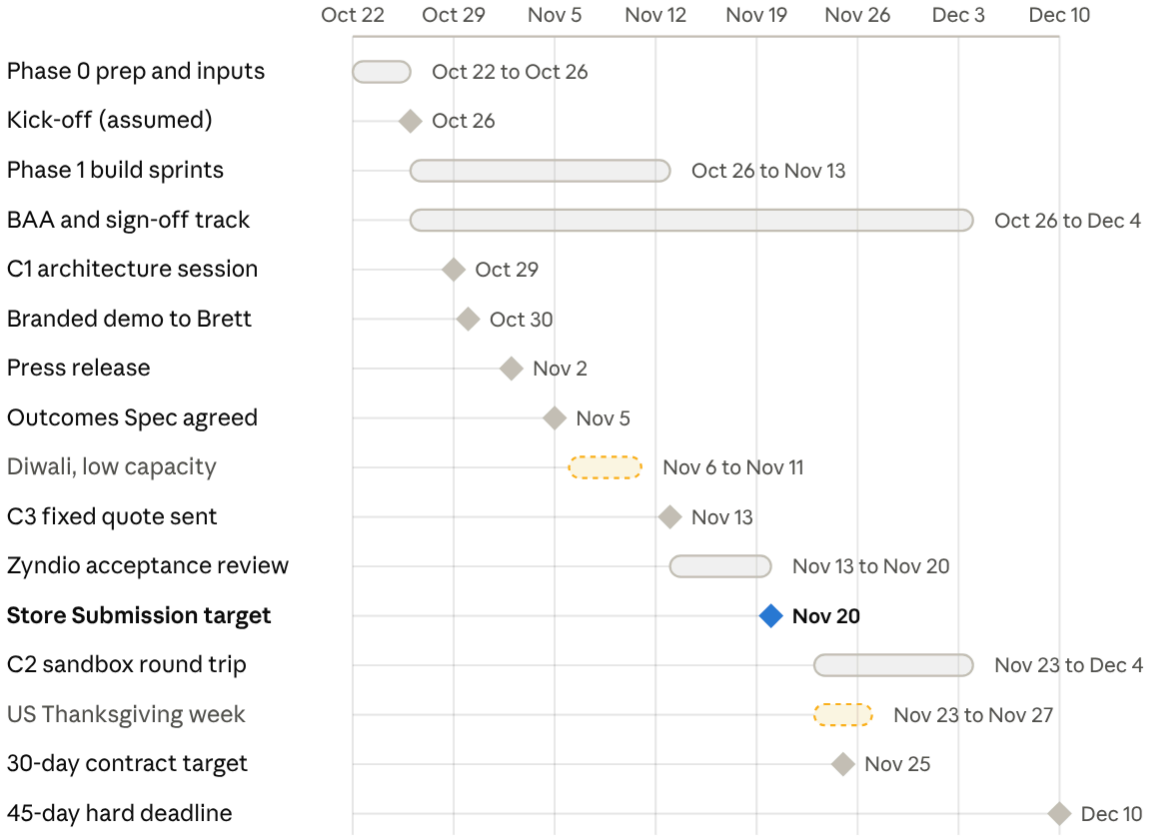
- Read-only access kept for 60 days after termination.
- Full patient data export in an agreed machine-readable format within 30 days of request: structured records as JSON, original documents as files, and a manifest.
- Deletion within 30 days after that, except legal or BAA holds, with a written deletion confirmation.

Delivery plan

If Zyndio's inputs arrive by Oct 26, we submit to both stores on Nov 20: five days ahead of the 30-day target and twenty days inside the hard limit, with Brett holding a branded demo before his Nov 2 release.

Store Submission lands Nov 20, three weeks inside the hard deadline

Plan by week, assuming Kick-off on Oct 26. Every build date moves day for day with Kick-off



delivery timeline · Oct 22 to Dec 10, 2026

The Nov 2 demo does not depend on Kick-off: it runs on fictional data through our own internal testing tracks, which is fine for Brett but must never be shown to DEXASCAN.

Sprint plan

Week	Dates	Goals	Exit check
W0	Oct 22 to 25	Inputs checklist to Zyndio, signed agreement confirmed, Zyndio tenant created, white-label build pipeline, golden fixture loaded, leak scan in CI	Inputs requested in writing; empty branded build compiles
W1	Oct 26 to Nov 1	Kick-off, C1 session (Oct 29), Zyndio branding, sign-in and consent, enrolment API skeleton	Branded demo build in Brett's hands Oct 30
W2	Nov 2 to 8	Apple Health and Health Connect, Health apps declaration, enrolment and status APIs with webhooks, portal roles. Outcomes engine starts in parallel	Outcomes Specification signed by Nov 5, before Diwali
W3	Nov 9 to 15 (Diwali, short week)	Portal configuration, companion boundaries, escalation contacts, treatment events API, store assets	C3 quote sent Nov 13; acceptance build to Zyndio Nov 13
W4	Nov 16 to 22	Zyndio review, fix list, store listings and privacy answers	Store Submission Nov 20
W5 to W6	Nov 23 to Dec 4	Store review responses, C2 round trip, review queue, delivery service, BAA and Compliance Sign-off	C2 success criteria pass; sign-off recorded
Buffer	Dec 5 to 10	Kept empty on purpose	45-day hard deadline is Dec 10

What happens after Dec 10

- **Billing starts** at store approval, likely late November, and no later than Dec 20 (30 days after a Nov 20 submission). Zyndio pays USD 399 a month from then even if real patients wait on the BAA, so Aditya should tell Brett this early.
- **Go-live gate** for the first real patient: BAA executed, Compliance Sign-off, Zyndio ranges and thresholds approved by its physician lead, escalation contacts live, on-call rota live, restore test and penetration test done.

- **C3** starts only on a signed Change Order. If it is signed in early December, production delivery can follow in January 2027, ahead of the first 90-day milestones for patients who start in December.
- **First real outcome records** arrive about 90 days after the first patients' baselines, so the 90-day review queue will first be used around March 2027.

Team, ownership and RACI

Aditya owns the relationship with Brett and every commercial answer; the engineering lead owns delivery and is the only person who commits dates to Zyndio's IT team. Names on our side are to be filled in at the internal kick-off.

Roles

Role	Who	Main responsibility
Commercial owner (AS)	Aditya Singh	Zyndio relationship, scope and price decisions, Change Orders, BAA signature
Engineering lead (EL)	To assign	This plan, architecture, dates, C1 session, acceptance, go-live gate
Mobile (MOB)	To assign	White-label app, Apple Health, Health Connect, store submissions
Backend (BE)	To assign	APIs, webhooks, identity, outcomes engine, delivery, metering
Data and AI (DATA)	To assign	Extraction, LOINC map, units, ranges, companion boundaries, safety behaviours
Web (WEB)	To assign	Physician portal configuration, review and unmatched queues, Partner Portal
QA (QA)	To assign	Test plans, golden fixtures, acceptance build, white-label leak checks
DevOps and security (OPS)	To assign	Environments, hosting, monitoring, on-call, security pack, subprocessor list

Role	Who	Main responsibility
Zyndio commercial	Brett Hudson	Approvals, inputs, DEXASCAN relationship
Zyndio technical	Matthew Trainer (to confirm as IT lead)	Zyndio Endpoint, zyndio.com changes, sandbox testing
Zyndio clinical	Physician lead, not yet named	Ranges, protocols, thresholds, narrative approval policy

RACI (R does it, A answers for it, C is consulted, I is kept informed)

Workstream	R	A	C	I
Zyndio Inputs and Kick-off	AS	AS	EL	Brett
White-label app and Store Submission	MOB	EL	QA, Matthew	AS, Brett
Wearables	MOB, BE	EL	DATA	AS
zyndio.com integration	BE	EL	Matthew	AS
Lab extraction, units and ranges	DATA	EL	Zyndio physician lead	AS
Physician portal configuration	WEB	EL	Zyndio physician lead	AS
Companion and safety behaviours	DATA	EL	Zyndio physician lead	AS
C1 session and Outcomes Specification	EL	AS	BE, DATA, Matthew, Zyndio physician lead	Brett
C2 sandbox round trip	BE	EL	Matthew, QA	AS, Brett
C3 fixed quote	EL	AS	BE	Brett
BAA and Compliance Sign-off	OPS	AS	EL, Zyndio counsel	Brett
Security pack and subprocessor list	OPS	EL	AS	Brett
On-call rota and SLAs	OPS	EL	all engineers	AS
Metering and invoices	BE	AS	OPS	Brett

Working rules with Zyndio

- One shared channel with Zyndio, agreed at kick-off (Schedule E). No side channels.
- Weekly written status from Aditya to Brett: done, next, blocked, and what we need from Zyndio.
- A decisions log kept in this doc. Anything Zyndio asks for outside Schedule A goes to Aditya before anyone says yes.
- Every wait on Zyndio, Apple or Google is logged with its start and end date. Those days extend our Store Submission window under clause 3.3, but only if we can show them.

Testing and acceptance

Zyndio accepts the build by approving Store Submission, or by letting 5 business days pass after we hand over the fictional-data build without reporting a material problem (clause 3.4). Our job is to make that review boring: everything below passes before the build goes to them.

Test layers

Layer	What it covers	Gate
Unit	Unit conversions, trend rules, completeness maths, HMAC tokens, signature checks	Every merge
Contract	JSON Schema for every API body, webhook and the outcome payload; Zyndio's sandbox calls replayed against it	Every merge
Extraction benchmark	Labelled set of fictional lab reports in the layouts US patients most often upload (national reference labs, hospital portals, phone photos). Field-level scoring of analyte, value, unit and date	Before acceptance build, and on every extractor change
Wearables	Real devices: iPhone with Apple Watch, an Android phone with a Wear OS or Fitbit or Samsung source. Backfill, background sync, revoked permissions, duplicate sources	Before acceptance build

Layer	What it covers	Gate
End to end	The golden fixture below, run through ingestion, record, outcome and delivery in sandbox	Before C2 sign-off
White-label leak scan	Automated search for "nostavia" (any case) across app binaries, web bundles, HTTP headers and bodies, emails, certificates, DNS targets and payloads	CI gate on every release; any hit blocks it
Security	Dependency, container and static scans in CI; access control tests per role; penetration test before the first real patient	Before go-live
Store readiness	Apple and Google review checklists, health data usage strings, Health apps declaration matching the manifest, in-app account deletion, privacy policy links	Before Store Submission
Accessibility	Dynamic type, screen reader labels and contrast on the core screens	Before acceptance build

Golden fixture (from the deck's example patient, fictional)

Every environment carries this patient. The outcomes engine must reproduce these numbers exactly.

Measure	Baseline	Follow-up	Expected change	Expected trend
Total weight	228 lb	194 lb	-34 lb, -14.9%	Improving
Fat mass	82 lb	55 lb	-27 lb, -32.9%	Improving
Lean mass	137 lb	133 lb	-4 lb, -2.9%	Depends on Zyndio's lean-loss threshold; must raise the flag the deck's narrative describes
HbA1c	6.2%	5.5%	-0.7 points, -11.3%	Improving

Measure	Baseline	Follow-up	Expected change	Expected trend
Fasting insulin	18.1 uIU/mL	9.4 uIU/mL	-8.7, -48.1%	Improving
ApoB	106 mg/dL	84 mg/dL	-22 mg/dL, -20.8%	Improving
hsCRP	3.8 mg/L	1.7 mg/L	-2.1 mg/L, -55.3%	Improving

C2 success criteria turned into tests (Schedule C)

- [ ] **Identity continuity:** the sandbox scan, the wearable data and the lab report all resolve to the same patient, and the outbound record carries the right partner_patient_id and zyndio_patient_token.
- [ ] **Traceable source data:** every value in the payload links back to a SourceDocument, a device or the scan record, with timestamp and unit.
- [ ] **Normalised biomarkers:** fixture values in mmol/L and IFCC units come out identical to the canonical-unit fixture.
- [ ] **Clean outbound record:** schema-valid, 25 to 35 fields, no raw wearable streams, no unapproved narrative, and zero Nostavia strings or identifiers.

Acceptance checklist for the build Zyndio reviews

- [ ] Zyndio branding on every screen, email and store asset; leak scan clean
- [ ] Sign-in, consent and onboarding with Zyndio's own wording (or clearly marked placeholders if not yet supplied)
- [ ] Lab upload and biomarker views working on the fixture set
- [ ] Apple Health and Health Connect connect, backfill and sync
- [ ] Physician portal roles, roster, timeline, brief and export working for Zyndio's named users
- [ ] Companion boundaries and emergency language screening tested with Zyndio's escalation contacts
- [ ] Enrolment and status APIs and webhooks working against Zyndio's sandbox
- [ ] Running on fictional data only, with Zyndio's default-range caveat visible until its own ranges are approved

Risks, open questions and decisions

The two risks that can hurt us most are commercial, not technical: Brett expecting a live product on Nov 2, and the C3 cap being set before anyone knows how DEXASCAN sends data. Both are handled in the next two weeks or not at all.

Risks, highest first

Risk	What happens if it lands	Mitigation	Owner
Nov 2 expectation gap. Brett said "3 weeks to get this done"; the contract window runs to late November	Disappointment at launch, pressure to cut corners or show real data	Aditya sends a short note this week: on Nov 2 Brett has a branded demo build on fictional data and an agreed outcomes design, which matches the release's own wording that nothing is operational yet	AS
C3 cap fixed before the data route is known	If DEXASCAN only has PDFs from many independent providers on different scanners, the connector costs more than USD 750 to build	In C1, push for one structured export from DEXASCAN through the Zyndio Endpoint. If PDFs are the only route, reuse our document pipeline and scope it to the scanner vendors in use. A second partner or cohort analytics is quoted separately (Schedule C)	EL
Signature mismatch. Brett says he signed via DocuSign; we sent Zoho Sign	Unclear Effective Date, weaker position if anything is disputed	Check Zoho Sign today, get the fully signed PDF, countersign if needed, record the Effective Date in this doc	AS

Risk	What happens if it lands	Mitigation	Owner
Developer accounts not ready	A new Google Play organisation account needs a D-U-N-S number, which can take up to 30 days; personal accounts must run a 14-day closed test with 12 testers first	Ask Zyndio today whether organisation accounts exist on both stores. If not, start enrolment immediately; Brett's demo can use internal testing in the meantime	AS, MOB
Zyndio cannot build the Zyndio Endpoint	C2 stalls, and C3 has nowhere to land	Offer, in C1, a relay we host on a Zyndio hostname. Nostavia stays invisible and Zyndio's IT effort drops to DNS and sign-off. Price it inside the C3 quote	EL
No source for GLP-1 treatment data	EHR integration is excluded, so dose and status may never reach us and the payload's treatment block stays empty	Treatment events API plus portal entry (POR-07). The Outcomes Specification names which fields are mandatory	EL
Health data store reviews	Health Connect declaration or HealthKit wording rejected, submission slips	Request only the data types we use, submit the declaration with the first upload, keep usage strings specific. Store-side delays extend our window under clause 3.3, so log them	MOB
Diwali week, Nov 6 to 11	Fewer working days mid-build in Lucknow	Outcomes Specification signed before Nov 6; no Zyndio-facing deadline set inside that week	EL
US Thanksgiving week, Nov 23 to 27	Slow approvals from Zyndio right after submission	Acceptance build to Zyndio by Nov 13 so their review closes before the holiday	EL

Risk	What happens if it lands	Mitigation	Owner
Wrong value extracted from an unfamiliar lab layout	A physician acts on a bad number	Per-field confidence, "needs review" state, source snippet beside every value, extraction benchmark gate	DATA
Mixed methods in trends (HRV SDNN against RMSSD, DEXA from different scanners)	A trend that looks clinical but is an artefact	Rules in the pipeline section: never merge, flag a method change as insufficient data	DATA
A Nostavia string leaks	Breach of clause 15.2 with Zyndio's partner	CI leak scan as a release blocker	QA
Usage invoice surprise	Background syncs count as "data processed" and push Active Patients up	Explain the definition to Zyndio before go-live and show the live count in the Partner Portal	AS
Payment rails	Razorpay failed three times; monthly invoices start at Billing Start	Use the Skydo route that worked for the onboarding fee, with wire as backup (clause 5.4)	AS

Open questions for Zyndio (most go into the C1 agenda)

- [ ] Do organisation Apple Developer and Google Play accounts exist in Zyndio's name? App name, bundle ID and store copy?
- [ ] Who is the IT lead (is it Matthew Trainer) and who is the physician lead?
- [ ] Which two standard modules does Zyndio want on Core? Do they want Re-Test Recall for the 90/180/365 reminders?
- [ ] Patient sign-in: our Zyndio-branded login, or federation with an identity provider Zyndio already runs?
- [ ] Which system holds Zyndio's clinical record, and can its backend send treatment events to our API?
- [ ] How does DEXASCAN deliver results today (API, file drop, PDF), how often, with which identifiers, and on which scanner vendors?
- [ ] Who builds and runs the Zyndio Endpoint: Zyndio, or a relay we host on Zyndio's hostname?

- [ ] Milestone anchor (treatment start or baseline scan) and the acceptable window around 90, 180 and 365 days
- [ ] Review policy: every outbound record clinician-approved, or numeric-only records released automatically when nothing is flagged?
- [ ] Minimum meaningful change per analyte and the lean-mass decline threshold (physician lead)
- [ ] Escalation contacts for emergency language and Priority 1 incidents
- [ ] Sender domain and DNS access for invites and notifications
- [ ] Partner disclosure consent wording (Zyndio's counsel) and retention period for the BAA
- [ ] Preferred device list (Schedule A, Part 3)
- [ ] Zyndio's legal entity name, state, address and notice email for the blank party block

Decisions needed from Aditya this week

1. Confirm the signed agreement and Effective Date, and countersign if it is still open.
2. Send Zyndio the inputs checklist above, today if possible, since Kick-off waits on it.
3. Set the Nov 2 expectation with Brett in writing.
4. Approve our C1 opening positions: Zyndio-delivered outbound, FHIR-aligned canonical record, clinician approval on every outbound record during the pilot, hosted relay offered as an option.
5. Approve staffing for the roles in the RACI and the on-call rota cost from go-live.
6. Decide whether to offer Re-Test Recall (USD 50 per month) or the Signature tier now, since the 90/180/365 re-scan loop is the use case that module is built for.

Appendix

A. Outcome payload, v1 (35 fields, 32 when the optional three are absent)

This is our proposal for the Outcomes Specification. Every measure uses the shared measure object described under the table. Meta fields (schema_version, record_id, record_version, generated_at, approved_at) sit in the envelope and are not counted.

#	Field	Type	Source
1	partner_patient_id	string	DEXASCAN, echoed
2	zyndio_patient_token	string	HMAC token, per partner

#	Field	Type	Source
3	milestone	enum: baseline, d90, d180, d365, adhoc	Milestone
4	baseline_date	date	Milestone
5	followup_date	date	Milestone
6	elapsed_treatment_days	integer	TreatmentCourse
7	medication	object: generic name, formulation	Zyndio
8	current_dose	object: value, unit	Zyndio
9	treatment_status	enum: active, paused, discontinued, completed	Zyndio
10	treatment_changes	array: type, date, clinician-approved label	Zyndio
11	weight	measure	Scan, wearable or clinic
12	bmi	measure	Derived
13	total_body_weight_change_pct	number	Derived
14	lean_mass_flag	enum: none, watch, flagged	Derived from scan and Zyndio threshold
15	hba1c	measure	Lab
16	fasting_glucose	measure	Lab
17	fasting_insulin	measure	Lab
18	triglycerides	measure	Lab
19	homa_ir (optional)	measure	Derived, same-draw rule
20	apob	measure	Lab
21	ldl_c	measure, with method (calculated or direct)	Lab
22	hdl_c	measure	Lab

#	Field	Type	Source
23	blood_pressure	object: systolic and diastolic measures, 30-day means	Wearable or clinic
24	resting_hr	measure, 30-day mean	Wearable
25	lp_a (optional)	measure, unit as reported	Lab
26	hscrp	measure	Lab
27	liver_enzymes	object: alt, ast, ggt measures	Lab
28	kidney_function	object: creatinine, egfr measures	Lab
29	activity_trend	trend object: steps per day, active minutes	Wearable
30	sleep_trend	trend object: duration, consistency	Wearable
31	hrv_trend	trend object with method (sdnn or rmssd)	Wearable
32	vo2max (optional)	measure	Wearable or partner site
33	data_quality	object: completeness by domain, confidence, provenance summary	Derived
34	clinician_summary	string, written or approved by a Zyndio clinician	Physician portal
35	next_measurement_milestone	object: milestone, due date	Milestone

Measure object: baseline and follow-up (each with value, unit, observed_at, source_type), abs_change, pct_change, trend (improving, stable, worsening, insufficient_data) and trend_reason. **Trend object:** window_days, baseline_mean, followup_mean, unit, coverage_pct, trend, and method where relevant.

B. Sample outcome.approved event (fictional golden fixture, shortened)

The remaining measures follow the same shape as hba1c.


```
{
  "event_id": "evt_01JD4K8Q2N",
  "event_type": "outcome.approved",
  "created_at": "2027-03-03T14:20:05Z",
  "schema_version": "1.0",
  "data": {
    "record_id": "rec_7HN2Q4",
    "record_version": 1,
    "approved_at": "2027-03-03T14:05:41Z",
    "partner_patient_id": "DX-88120",
    "zyndio_patient_token": "zpt_9c4f1e7a2b",
    "milestone": "d90",
    "baseline_date": "2026-12-01",
    "followup_date": "2027-03-01",
    "elapsed_treatment_days": 90,
    "treatment_status": "active",
    "weight": {
      "baseline": {"value": 228, "unit": "lb", "observed_at": "2026-12-01",
"source_type": "partner_scan"},
      "followup": {"value": 194, "unit": "lb", "observed_at": "2027-03-01",
"source_type": "partner_scan"},
      "abs_change": -34,
      "pct_change": -14.9,
      "trend": "improving",
      "trend_reason": "change exceeds configured minimum"
    },
    "total_body_weight_change_pct": -14.9,
    "lean_mass_flag": "watch",
    "hba1c": {
      "baseline": {"value": 6.2, "unit": "%", "observed_at": "2026-11-28",
"source_type": "lab"},
      "followup": {"value": 5.5, "unit": "%", "observed_at": "2027-02-26",
"source_type": "lab"},
      "abs_change": -0.7,
      "pct_change": -11.3,
      "trend": "improving",
      "trend_reason": "change exceeds configured minimum"
    },
    "data_quality": {"completeness": {"labs": 0.92, "wearables": 0.81,
"scan": 1.0}, "confidence": "high"},
    "clinician_summary": "Weight and adiposity decreased with improvement in
available cardiometabolic biomarkers. Lean mass also declined and remains an
area for clinical attention.",
```

```
    "next_measurement_milestone": {"milestone": "d180", "due_date": "2027-05-
30"}
  }
}
```

C. LOINC starting map for the priority analytes

The data lead verifies every code against the current LOINC release before it ships.

Analyte	LOINC
HbA1c	4548-4
Glucose, fasting	1558-6
Insulin	20448-7
Triglycerides	2571-8
Cholesterol, total	2093-3
LDL-C, calculated / direct	13457-7 / 18262-6
HDL-C	2085-9
Non-HDL-C	43396-1
ApoB	1884-6
Lp(a), mass	10835-7 (molar results map to a separate code)
hsCRP	30522-7
ALT / AST / GGT	1742-6 / 1920-8 / 2324-2
Creatinine	2160-0
eGFR, CKD-EPI 2021	98979-8
TSH / Free T4 / Free T3	3016-3 / 3024-7 / 3051-0
Testosterone, total / Estradiol / SHBG	2986-8 / 2243-4 / 13967-5
25-OH vitamin D (D2+D3)	62292-8
Ferritin / IGF-1	2276-4 / 2484-4
Body weight / BMI	29463-7 / 39156-5
Blood pressure systolic / diastolic	8480-6 / 8462-4
Resting heart rate	40443-4

D. Glossary

Term	Meaning in this plan
Kick-off Date	Latest of: last signature, fee received, Zyndio Inputs received. Starts the build window
Store Submission	Our submission of the production build for Apple or Google review under Zyndio's accounts
Billing Start Date	Earlier of store approval and 30 days after first Store Submission
Active Patient	Signed in, or had data processed, in a calendar month
Compliance Sign-off	Written confirmation by both parties that the Schedule D assessment is complete
Zyndio Endpoint	Zyndio's API endpoint that receives DEXASCAN data before passing it to us
Outcomes Specification	The written output of C1, agreed by both parties; defines fields, rules and routes
Change Order	Signed document for any work outside Schedule A, with fixed fees
Minimum meaningful change	Per-analyte threshold, set by Zyndio's physicians, that separates a real change from noise

E. Sources

Internal: Platform Licence Agreement (revised Oct 2026), Zyndio strategy deck, Zyndio and DEXASCAN press release, LinkedIn thread with Brett Hudson. External, checked Oct 22, 2026:

- Publish your health app on Google Play (Health apps declaration)
- Health Connect: read raw data (30-day default, history and background permissions)
- Google Play declaration of Health Connect access
- Configure Google Play closed testing (12 testers, 14 days, organisation accounts exempt)
- Google Play 12-tester rule (D-U-N-S can take up to 30 days)
- Diwali 2026 calendar (Nov 6 to 11)